



VoiceCare AI — Post-Discharge Patient Monitoring

WHY

Explain the Problem

Problem Description & Business Scenario :30-day hospital readmissions cost the Indian healthcare system ₹15,000–₹80,000 per patient. Most readmissions happen silently — patients go home, miss medications, misunderstand discharge instructions, and return in crisis. No scalable follow-up system exists for non-English, non-app users.

Problem Scope :Affects 18–22% of discharged patients across tier-1 and tier-2 hospitals in India. Particularly severe in cardiac, diabetic, and post-surgical cases. Overburdened nursing staff cannot manually call hundreds of patients daily.

Target Users/Stakeholders: Discharged patients (Tamil/Hindi/Telugu/Kannada/Malayalam speakers), nursing staff, treating doctors, hospital administrators, and insurance providers.

Solution Overview : An AI-powered voice agent that auto-calls discharged patients daily in their native language — no app, no smartphone needed. It holds natural, empathetic conversations and detects early warning signs before they become emergencies.

Technical Details :Multilingual voice AI (IVR + LLM) with real-time speech analysis. Three-signal risk engine: (1) Vocal biomarkers — tiredness, breathlessness, hesitation; (2) NLP keyword detection — negative phrases, confusion signals; (3) Symptom reporting — fever, chest pain, dizziness. Risk-based automation: WhatsApp alerts → nurse notification → doctor video call scheduling.

Innovation : Works on ANY phone (no app needed). Detects risk from *how* a patient speaks, not just *what* they say. First multilingual post-discharge voice AI in Indian regional languages.

Market Potential : 50M+ hospital discharges/year in India. \$2.4B readmission prevention market. Scalable to insurance, home healthcare, and pharmacy follow-up segments.

How

Explain the Solve

WHAT

Value proposition

Primary Benefits :Daily automated patient follow-up, zero marginal cost, native language.

Efficiency & Flexibility :Nurses attend only high-risk cases. 100s of simultaneous calls. 24/7, no staff fatigue.

Time and Cost Saving : 70% less nurse call time. 45%+ fewer readmissions. ₹12K–₹50K saved per avoided case

Scalability: Any phone, any hospital. No hardware needed. Multi-hospital dashboard ready.

Social Impact : Bridges health literacy gap. Equal care across languages & income groups.



Financials & Timelines | Business Plan (2/2)

Investments

What does it take & How much does it cost to solve?

Voice AI + Multilingual NLP	LLM fine-tuning for Tamil, Hindi, Telugu, Kannada, Malayalam	₹40–60L
Vocal Biomarker Engine	Cloud hosting, IVR integration, WhatsApp API	₹8–12L/year
IVR + WhatsApp Integration	2 AI engineers, 1 clinical validator, 1 ops lead	₹25–35L/year
Doctor Dashboard	Daily risk dashboard + audio snippet viewer	₹15–20L

Reduction in 30-Day Readmissions	45%+ drop	via early vocal + symptom risk detection
Nurse Follow-Up Time Saved	70% faster	triage — only high-risk cases escalated
Cost Saved per Patient	₹12,000–₹50,000	saved per avoided readmission
Missed Medication Detection	Real-time	WhatsApp alerts to family + nursing staff

Returns

Quantify the benefits & What if I don't solve?

Timelines

Time to realize benefits

Phase 1: AI Build & Language Training	Weeks 1–2	Voice model + 5-language NLP + risk engine setup
Phase 2: Pilot Launch	Weeks 3	3 hospitals, 500 patients — live call testing
Phase 3: Dashboard & Alerts	Weeks 4–5	Doctor dashboard + WhatsApp + video call integration
Phase 4: Full Deployment	Weeks 6–10	Scale to 20+ hospitals — ROI realized, readmissions tracked